

IOS-XR Nettools MCP

User Guide for the authenticated, read-only network inspection service

Service status: the MCP service runs in a persistent Docker container on a4000 and exposes native MCP streamable HTTP directly on port 8000. The host publishes plain HTTP on all interfaces for trusted LAN/VPN clients.

What This Service Does

The service exposes reviewed Cisco IOS-XR lab inspection tools through the Model Context Protocol (MCP). It can inspect inventory, interface and routing protocol state, deterministic device health, tickets, logs, metrics, and selected topology sources.

- Device commands are reconstructed from typed inputs and an allowlist.
- There is no arbitrary command tool, configuration mode, shell access, or network remediation capability.
- Raw device command text is sanitized before MCP clients receive results.
- Active probes are disabled by default.

Connect an MCP Client

1. Use the direct HTTP endpoint

Replace `<a4000-host-or-ip>` with the host name or LAN/VPN address reachable from the client:

```
http://<a4000-host-or-ip>:8000/mcp
```

2. Configure an MCP client

Use a client that supports MCP streamable HTTP. Supply the bearer token through the client's secret store or a local environment variable. Do not place the token in a committed configuration file.

```
{
  "mcpServers": {
    "ios-xr-nettools": {
      "url": "http://<a4000-host-or-ip>:8000/mcp",
      "headers": {
        "Authorization": "Bearer ${IOS_XR_NETTOOLS_MCP_TOKEN}"
      }
    }
  }
}
```

Client configuration formats differ. The essential requirements are the endpoint URL and an `Authorization: Bearer <token>` header on every MCP request.

Authentication and Access

Property	Current behavior
Transport	Native MCP streamable HTTP
Authentication	Shared bearer token; missing or invalid tokens receive HTTP 401
Encryption	None. Traffic and bearer tokens travel over plain HTTP.
Network exposure	All host interfaces on TCP port 8000 by default (0.0.0.0:8000)
Authorization	All valid token holders receive the same read surface; there is no per-user RBAC

Use this endpoint only on a trusted LAN or VPN; never expose it directly to the public internet. It does not provide TLS termination, per-user identity, token rotation, RBAC, or rate limiting. Plain HTTP does not protect the bearer token from network observers.

Useful First Calls

Tool	Use
<code>list_lab_devices</code>	List the lab inventory.
<code>check_lab_interfaces</code>	Inspect interface status for a device.
<code>check_lab_bgp_neighbors</code>	Inspect BGP neighbor state.
<code>check_lab_isis_neighbors</code>	Inspect IS-IS adjacency state.
<code>assess_lab_device_health</code>	Run deterministic health rules and return findings.
<code>investigate_lab_session</code>	Follow a deterministic dependency ladder for a BGP, interface, IS-IS, or LDP subject.
<code>list_lab_tickets</code>	List open tickets. Set <code>include_closed=true</code> to include completed investigations.

Operational Notes

- The service uses strict SSH host-key verification for routers.
- Tickets, evidence, admission state, and event state are persisted in a Docker named volume and survive container restart.

- The service restarts automatically after a host reboot through Docker Compose.
- NetBox and Neo4j tools use Docker service DNS when those backing services are running. If either backing service is down, its MCP tool returns a structured error.
- External-source tools depend on the availability of their respective platform services.

Support Checklist

1. Confirm the client can reach `<host-or-ip>:8000` over the trusted LAN/VPN.
2. Confirm the MCP client sends the bearer token.
3. Call `list_lab_devices` before a device-specific tool.
4. For a returned object refusal, refresh the relevant device check and verify the requested object exists.
5. For NetBox or graph errors, verify the corresponding backing service is running before changing MCP configuration.

IOS-XR Nettools MCP User Guide. This guide intentionally omits bearer tokens, router credentials, and internal secret values.